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ORIGINAL ARTICLE

Chin Tuck Against Resistance (CTAR): New Method for Enhancing Suprahyoid Muscle Activity Using a Shaker-type Exercise

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Abstract For patients with dysphagia resulting from upper esophageal sphincter dysfunction, strengthening the suprahyoid muscles through therapeutic exercise has proved effective in restoring oral feeding. The aim of this study was to compare the maximum and mean surface electromyography (sEMG) activity of the suprahyoid muscles during the Chin Tuck Against Resistance (CTAR) exercise and the Shaker exercise for both isokinetic and isometric tasks. During the CTAR exercises, the participant is seated while tucking the chin to compress an inflatable rubber ball, whereas during the Shaker exercise, the participant is lay supine while lifting the head to look at the feet. Forty healthy participants (20 males, 20 females) aged 21–39 years completed all four tasks in counterbalanced order, with measures of resting activation taken prior to each exercise. Although subjective feedback suggested that the sitting position for CTAR is less strenuous than the supine position for Shaker, the results of separate analyses showed significantly greater maximum sEMG values during the CTAR isokinetic and isometric exercises than during the equivalent Shaker exercises, and significantly greater mean sEMG values were observed for the CTAR isometric exercise than for the Shaker isometric exercise. Clinical trials are now needed, but the CTAR exercises appear effective in exercising the suprahyoid muscles, and they could achieve therapeutic effects comparable to those of Shaker exercises, with the potential for greater compliance by patients.

Keywords Deglutition disorders · Deglutition · Dysphagia exercises · Surface electromyography

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For patients with pharyngeal dysphagia involving incomplete upper esophageal sphincter (UES) opening, therapeutic exercises to strengthen the muscles that help open the UES can improve swallowing ability. The suprahyoid group of muscles facilitates opening the UES [1, 2]. The participation of this muscle group is enhanced by performing head lifts in a supine position [3] with the patient instructed to look at their toes. This exercise is commonly known as the Shaker exercise.

Shaker exercise consists of sustained head lifts (isometric) and successive head lifts (isokinetic), performed by the patient three times a day while in the supine position [4]. Exercising the suprahyoid muscles is the main goal of the Shaker exercise [4]. The contraction of the suprahyoid muscles contributes to the upward and forward movement of the larynx and hyoid bone, resulting in opening of the UES [5]. The strong correlation between the activation of the suprahyoid muscles and the superior–anterior movement of the hyoid bone [6] supports the mechanical role of the suprahyoid muscles in elevating the hyoid and opening the UES. The Shaker exercise was developed to strengthen the suprahyoid muscles and thereby help increase the opening of the UES [2, 7]. Strengthening the suprahyoid muscles by performing the Shaker exercise has been found to be effective in restoring oral feeding in patients with pharyngeal dysphagia due to incomplete UES opening [4].

Research has shown that performing the Shaker exercise significantly increased the anteroposterior diameter of the UES in elderly patients with dysphagia and older adults without dysphagia [7, 8], and the patients with dysphagia exhibited a significant reduction in post-swallow aspiration [2]. This effect is attributed to improved strength of the suprahyoid muscles after completion of the Shaker exercise regimen [4, 9]. Findings from spectral analysis of sEMG measurements taken while the patient performed the Shaker exercise provided evidence that the suprahyoid muscles showed signs of fatigue, indicating that they are physiologically affected by the exercise [3]. Muscles that are affected during regular exercise would therefore be strengthened upon completion of the exercise goals.

Although supervision is not required for the Shaker exercise, compliance may be a problem. Easterling et al. [8] concluded that a structured and gradually progressive program is necessary for the elderly to follow to achieve the Shaker exercise goals. In their study of 26 older adults (aged 66–93 years) with no current swallowing problems, only 50 % of the participants completed the prescribed isometric goals and only 70 % completed the prescribed isokinetic goals in an exercise regimen that spanned 6 weeks. Muscle discomfort and time constraints were the main reasons participants gave for failure to attain the exercise goals. Also, given that more participants attained the isokinetic exercise goal every week compared to the isometric exercise goal [8], it suggests that the isometric Shaker exercise may be especially challenging.

In addition, Yoshida et al. [10] reported that performing head lifts in the supine position may be too demanding physically for elderly patients with chronic disease. This is supported by anecdotal feedback from some of our colleagues. Many patients with dysphagia have accompanying conditions (e.g., chronic cough, pneumonia, heart disease, malnutrition) that contribute to the challenges they face in performing therapeutic exercises, even when they are highly motivated. This suggests that the development of therapeutic exercises that are less strenuous than Shaker but equally effective in exercising the suprahyoid muscles would improve patient compliance, especially by those who face significant challenges in performing physical therapy. One possibility is the Chin Tuck Against Resistance (CTAR) exercise, which can be performed with the patient seated in a chair. The resistance in the Shaker exercise is lifting one's head against gravity. For the CTAR exercise, the resistance is compressing an inflatable rubber ball that is placed between the patient's chin and the manubrium sterni. The patient is required to tuck the chin toward the manubrium sterni while squeezing the ball.

Chin tuck is already a widely used compensatory strategy in dysphagia management. Based on a questionnaire survey of speech language pathologists in Japan and the US, chin tuck is perceived as head flexion or neck flexion or a combination of both [11]. Both head flexion [12] and neck flexion [13, 14] are known to activate the suprahyoid muscles [12, 14], and the combined contraction of both suprahyoid and infrahyoid muscles will pull the mandible toward the sternum. The trajectory of the head and neck flexion in the CTAR exercise will mirror that in the Shaker exercise even though the CTAR exercise is performed when seated rather than when lying down. Therefore, chin tuck against resistance, in the form of a rubber ball, should strengthen the suprahyoid muscles in much the same way as the Shaker exercise does.

The impact of the CTAR exercise on the suprahyoid muscles needs to be tested on normal adults before it can be used on patients with dysphagia. Thus, the aim of the present study was to find out if the CTAR exercise is as effective as the Shaker exercise in raising the sEMG activation levels of the suprahyoid muscles during both isometric and isokinetic tasks. If greater (or equivalent) sEMG activity of the suprahyoid muscles is observed during the CTAR exercise, it could be used as an alternative therapeutic intervention for patients who find the Shaker exercise in the supine position physically challenging.

Materials and Methods

Design

The design was repeated measures with the order of the two exercises (CTAR and Shaker) and the two tasks (isometric and isokinetic) counterbalanced across participants within gender. Resting activation levels were recorded immediately prior to each of the four test trials.

Participants

Forty adult participants (20 males and 20 females) ranging in age from 21 to 39 years (mean [M] = 29.13, standard deviation [SD] = 5.694) were recruited. All participants completed a health questionnaire and gave informed consent stating that they were healthy and had no history of swallowing problems, physical disability of the upper body, and/or neurological disease. They were also screened for adequate oromotor structure and functioning by a trained speech-language pathologist.

Procedure

Each participant was evaluated in a quiet room. The therapeutic benefits of the exercises were explained to the participants before they signed the informed consent form and completed the health questionnaire. The screening

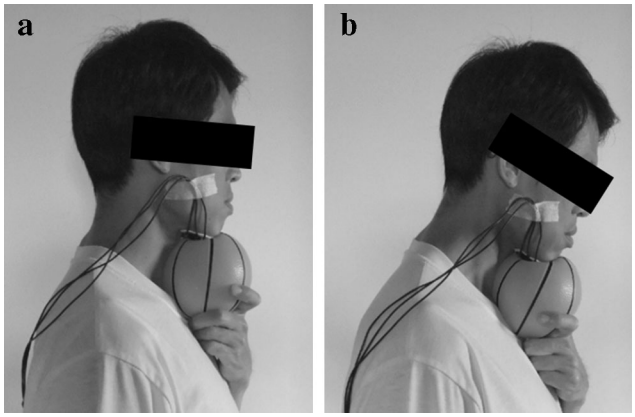

Fig. 1 Example of the CTAR exercise with the electrodes patch attached. **a** At rest position. **b** Chin tuck to compress a rubber ball

procedure ended with an oromotor examination. This was followed by a brief explanation of the four exercise tasks and a short demonstration of the CTAR exercise and the Shaker exercise by the experimenter.

Every participant performed one trial for each of the four exercise tasks: (1) CTAR isometric, (2) CTAR isokinetic, (3) Shaker isometric, and (4) Shaker isokinetic. A 5-min rest period was provided between the tasks. The mean and the maximum sEMG resting baselines were recorded prior to each exercise. The participants were also asked if they felt any discomfort before proceeding with the next trial. The isometric task was performed for a total of 10 s while the isokinetic task was 10 successive repetitions. The 10 s duration and the 10 repetitions were adapted from the study by Yoshida et al. [10]. Ten seconds is enough time to obtain a stable sEMG measurement during the utilization of the suprahyoid muscles for comparison purposes, but it minimizes the possibility of undue muscle fatigue that might affect the performance of subsequent tasks. The participants were reminded to breathe and to keep their mouths closed during the Shaker exercise [8] and the CTAR exercise.

For the Shaker exercise, the participants laid flat on their back (supine position) on an exercise mat (a 5-mm-thick "yoga" foam mat) to perform two tasks: (1) lift and hold the head up for 10 s (isometric), and (2) lift the head ten times successively (isokinetic). The experimenter counted aloud each completed head lift. The participants were reminded to lift the head high enough such that they could see their own feet without raising their shoulders. For the CTAR exercise, the participants were seated on a chair with their back upright to perform two tasks: (1) squeeze the rubber ball that was placed between the chin and the manubrium sterni by tucking the chin as hard as possible and sustaining it for 10 s (isometric), and (2) squeeze the rubber ball by tucking the chin as hard as possible ten successive times (isokinetic). The experimenter counted aloud each completed chin tuck. The participants were allowed to use one hand to keep the ball in position during the exercise. Also, they were reminded to keep their shoulders stationary and to keep the chin in contact with the ball throughout the exercise (Fig. 1). The same inflatable rubber ball, ~12 cm in diameter, was used by all participants.

Subjective Feedback

After a 5-min rest after completion of all four exercise tasks, each participant was then asked verbally which of the two exercises was less strenuous, and their binary responses were recorded in the final section of the health questionnaire.

sEMG Data Collection

The sEMG data were collected using the MyoTrac Infiniti™ (Thought Technology Limited, Montreal, QC, Canada) encoder during all the exercise tasks. After cleaning the skin surface of the suprahyoid region with an alcohol swab, a single-use electrode patch (T3402M EMG Triode™ Electrode, Thought Technology Limited) with adhesive backing was attached to each participant's suprahyoid region (area between the chin and hyoid bone). Conductive electrode gel (signa gel®, Parker Laboratories Inc., Fairfield, NJ, USA) was used to improve conduction. Each electrode patch contained three electrodes (two active and one reference) in a triangular configuration and was connected to the encoder via an EMG extender cable (T8710M, Thought Technology Limited). The two active electrodes were positioned such that they were aligned along the midline (sagittal plane) of the suprahyoid area.

The recorded sEMG data were analyzed offline. For each exercise task, the onset and offset representing the effort by the participant for each exercise task were identified and the signals in-between were analyzed to obtain the mean and maximum values for each participant. For the isometric task, the onset was defined as the point where the sEMG signal began to level off (after it had risen), and the offset was the point before the sEMG signal dropped rapidly to baseline level. For the isokinetic task, the onset was the point at baseline level just before the sEMG signal began to rise rapidly at the start of the first repetition, and the offset was the point where the sEMG signal reached baseline level at the end of the tenth repetition. There were two sEMG measures per exercise task: mean and maximum sEMG values. These two sEMG values were depicted in microvolt root mean square (μV_{RMS}) and obtained using the statistics function in the software provided with the encoder (BioGraph Infiniti, Thought Technologies Limited). A total of eight sEMG measures (mean and

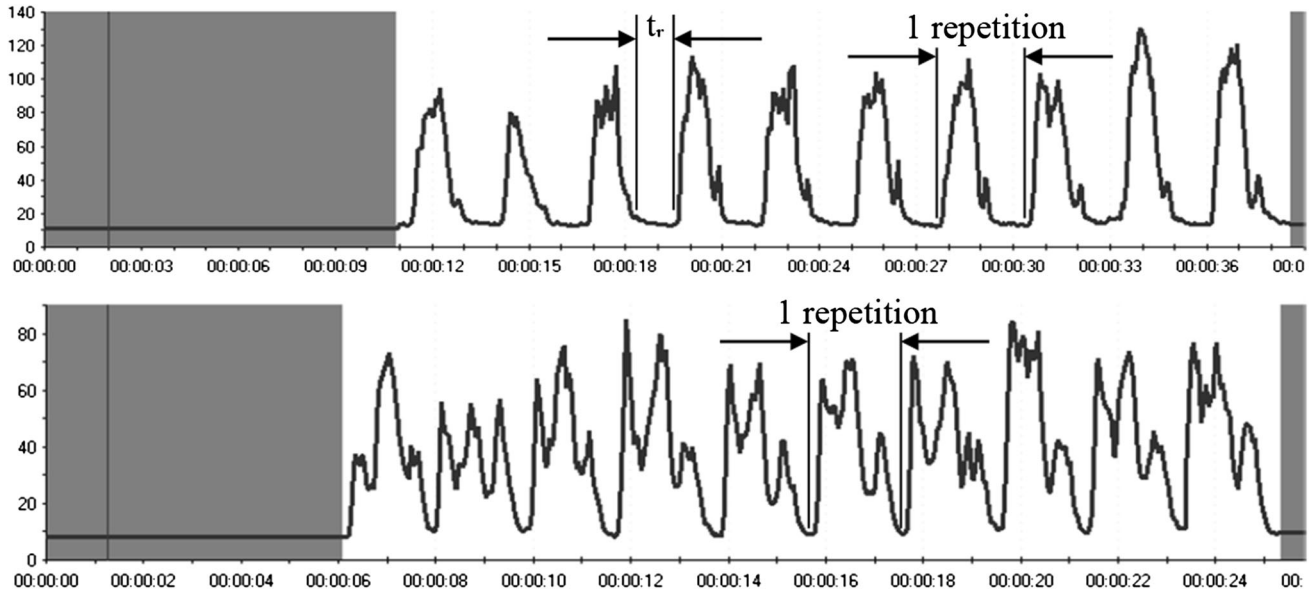

Fig. 2 sEMG plots during isokinetic task. *Top* CTAR. *Bottom* Shaker

maximum for both the isometric and isokinetic tasks for CTAR and Shaker) and four resting-level recordings were obtained for each participant. An example of the recording for a single participant during the isokinetic task for the CTAR and Shaker exercises is shown in Fig. 2.

Statistical Analysis

To ascertain whether there were significant differences between the CTAR and Shaker exercises for both the isometric and isokinetic tasks, the maximum and mean sEMG measures for the 40 adult participants were compared in separate 2 (Exercise Type: CTAR/Shaker) × 2 (Task Type: isometric/isokinetic) repeated-measures ANOVAs using SPSS version 21 (SPSS, Inc., Chicago, IL, USA), with Bonferroni corrections for simple-effects testing.

Results

The order of the four exercises was counterbalanced within gender and across all participants to minimize undue effects of fatigue. The resting baselines prior to each exercise were as follows: CTAR Max M = 16.97, SD = 7.80; CTAR Mean M = 15.07, SD = 7.79; Shaker Max M = 19.94, SD = 10.94; Shaker Mean M = 18.97, SD = 11.05. The data suggest that the resting baselines were similar for the four exercises with the ball in place, and none of the participants reported undue discomfort during any of the trials. The means and standard deviations for the maximum and mean sEMG activation levels during the four CTAR and Shaker exercises are given in Table 1.

Table 1 Maximum and mean sEMG values of isometric and isokinetic tasks obtained during the CTAR exercise and the Shaker exercise

sEMG measure/task	CTAR mean (SD) sEMG	Shaker mean (SD) sEMG
Maximum/isokinetic	205.83 (114.06)	154.23 (84.56)
Mean/isokinetic	57.13 (30.23)	55.60 (25.41)
Maximum/isometric	166.52 (99.62)	65.91 (35.70)
Mean/isometric	103.72 (64.03)	42.95 (21.70)

Comparison of Maximum Activation Levels for CTAR and Shaker Exercises

The results of the 2 × 2 repeated-measures ANOVA showed that the CTAR exercise resulted in significantly greater maximum activation than the Shaker exercise [$F(1, 39) = 17.242$, MSE = 231,659.05, $p < 0.001$, $\eta^2_{\text{partial}} = 0.307$] and that the isokinetic task resulted in significantly greater activation than the isometric task [$F(1, 39) = 43.193$, MSE = 162,915.87, $p < 0.001$, $\eta^2_{\text{partial}} = 0.526$]. There was also significant interaction between exercise type and task [$F(1, 39) = 8.264$, MSE = 24,020.29, $p = 0.007$, $\eta^2_{\text{partial}} = 0.175$], with a greater advantage of isokinetic over isometric for the Shaker exercise ($p < 0.0125$, $d = 1.361$) than for the CTAR exercise ($p < 0.0125$, $d = 0.367$).

Comparison of Mean Activation Levels for CTAR and Shaker Exercises

The results of the second 2 × 2 repeated measures ANOVA also showed that the CTAR exercise resulted in

significantly greater mean sEMG activation levels than the Shaker exercise [$F(1, 39) = 14.518$, MSE = 38,813.21, $p < 0.001$, $\eta^2_{\text{partial}} = 0.271$], but that the isometric task resulted in significantly greater activation than the isokinetic task [$F(1, 39) = 20.163$, MSE = 11,520.42, $p < 0.001$, $\eta^2_{\text{partial}} = 0.341$]. For mean sEMG, the interaction between exercise type and task type was also significant [$F(1, 39) = 74.181$, MSE = 35,101.81, $p < 0.001$, $\eta^2_{\text{partial}} = 0.655$], with an advantage for isometric over isokinetic for CTAR ($p < 0.0125$, $d = 0.931$) and an advantage for isokinetic over isometric for Shaker ($p < 0.0125$, $d = 0.535$).

Gender Effects

Gender was not a main variable of interest in this study but we recruited an equal number of males and females in the same age range and then checked whether the results were consistent for the CTAR and Shaker isokinetic and isometric exercises. The mixed 2 × 2 × 2 ANOVAs (CTAR/Shaker × isokinetic/isometric × males/females) for mean and maximum sEMG activation levels confirmed that there were no significant main effects of gender, nor did gender interact with exercise type or movement type (all $p > 0.05$).

Feedback from Participants

From the subjective feedback based on a binary decision, 80 % of the participants (16 males and 16 females) reported that the CTAR exercise was less strenuous than the Shaker exercise.

Discussion

The aim of this study was to ascertain whether the CTAR exercise is as effective as the Shaker exercise in raising the sEMG activation levels of the suprahyoid muscles during both isometric and isokinetic tasks. The maximum and mean values of the sEMG measurements for each task were recorded and analyzed, and participants were also asked to report which exercise was less strenuous. Resting baseline sEMG measures taken prior to the four exercises were found to be similar, order was counterbalanced across the participants, and there were no gender effects.

The sEMG results for the maximum activation levels showed that the CTAR exercise, using a ball as resistance under the participant's chin, resulted in significantly greater activation during both the isokinetic and isometric tasks. The CTAR exercise also resulted in significantly greater activation than did the Shaker exercise for the mean sEMG measure for the isometric task but not for the isokinetic task. These findings demonstrate that the CTAR exercise does have an equivalent or greater impact than the Shaker exercise on the suprahyoid muscles, even though it was reported as less strenuous. One reason for the higher sEMG levels during CTAR might be that the ball itself stretches the suprahyoid muscles prior to both the isometric or isokinetic tasks. When a muscle is lengthened during stretch, nerve activity is increased and the muscle fibers contract (myotactic reflex) to resist stretching. So although the resting sEMG levels for CTAR and Shaker were equivalent, the pressure of the ball may enhance activation regardless of the task. This does not negate the advantage of CTAR over Shaker for improving suprahyoid muscle strength but suggests that the ball's size and consistency might also be important factors.

Interestingly, the simple effects for the mean sEMG revealed greater activation for the isometric than for the isokinetic task for the CTAR exercise, but greater activation for the isokinetic than for the isometric task for the Shaker exercise. This was surprising given that the Shaker isometric exercise was reported to be especially challenging for patients [8]. One possible reason for this might be related to the amount of effort required of the participant to return the head to the resting position. For the Shaker exercise, the participants performed head lifts while lying on an exercise mat on the floor. Effort is required of the participants when lifting the head, but it was noted that considerable effort was needed to lower the head back onto the mat and into the resting position. With the CTAR exercise, however, effort was required to tuck the chin to compress the rubber ball but not in releasing the compression: the inflatable ball expands back to its original shape, thereby assisting the participant in moving the chin back to the resting position with minimal or negligible effort. This was apparent from the sEMG plot for the CTAR isokinetic task (Fig. 2). From the plots it was noted that approximately half the time (t_r) of each repetition was spent at the resting baseline, resulting in a smaller area under the graph when compared to the sEMG plot for the Shaker isokinetic task.

One possible way to increase the duration of the effort during the CTAR isokinetic exercise would be to instruct the participant to release the compression of the rubber ball gradually and in a controlled manner. This would translate to a slower rate of decline in the sEMG value instead of a steep drop to the baseline value seen in Fig. 2, and then an increase in the area under the curve and hence a greater mean sEMG value. Alternatively, the number of repetitions can be increased to produce more activity in the suprahyoid muscles.

Clinical trials are now needed to test these ideas and the basic utility of CTAR. However, the CTAR exercises appear effective in exercising the suprahyoid muscles in healthy

participants, and they could achieve therapeutic effects comparable to those of Shaker for strengthening the suprahyoid muscles and thereby increasing the anteroposterior diameter of a patient's UES [4, 8]. CTAR also has the potential to increase compliance in patients who find the supine position required for the Shaker exercise inconvenient or uncomfortable. Compliance is an important issue for clinicians in the rehabilitation of elderly patients with dysphagia. Previous sEMG studies on Shaker exercise have shown that the sternocleidomastoid muscle fatigued before the suprahyoid muscles during the isometric and isokinetic exercises [3, 15]. The fatigue of the sternocleidomastoid muscle prior to the suprahyoid muscles might limit the performance of the Shaker exercise and attainment of the exercise goals [8]. This in turn may result in an increased likelihood of (sternocleidomastoid) muscle discomfort due to inadequate muscle strength in elderly patients if they put in excessive effort when performing head lifts. Muscle discomfort has been one of the main reasons for noncompliance and not reaching the exercise goals [8]. Thus, the relative ease perceived by the participants in performing the CTAR exercise over the Shaker exercise may be due to the reduced involvement of the sternocleidomastoid muscle as lifting the load of one's head is eliminated. During the CTAR exercise, the rubber ball is squeezed using one's best effort. This allows elderly patients to perform the CTAR exercise based on their current strength level without having to be strong enough to lift their head from the supine position during each session. Being able to control exertion level on a day-to-day basis could contribute to improved compliance and attainment of the exercise goals that have been set by the clinician. In addition, the CTAR exercise has the relative advantage of convenience, especially for patients who are actively mobile and like to spend time outside the home.

The results obtained with the CTAR exercise are encouraging but two limitations of this study need to be acknowledged. First, the data were derived from only one trial of each of the four exercises. To minimize possible confounding from undue fatigue, care was taken to counterbalance task order across participants and resting baselines were recorded. In future studies, however, more extensive testing over several trials for each task, ideally on separate occasions with measures of inter-rater reliability, should be considered. The second, and more obvious, limitation of this study is that it was conducted with healthy young adults who were able to complete CTAR and Shaker with relative ease. Replication with older adults, prior to clinical studies with patients with dysphagia, would increase our understanding of the impact of aging, such as variability in fitness level and strength, as well as the ability to follow the exercise instructions.

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